

LIDAR Inspection Report of MNB ch.1039+350

Bridge No: MNB - 1040/2

Bridge Chainage: 1039+350 (CKRP)

Latitude/Longitude: 25.08735, 75.64816



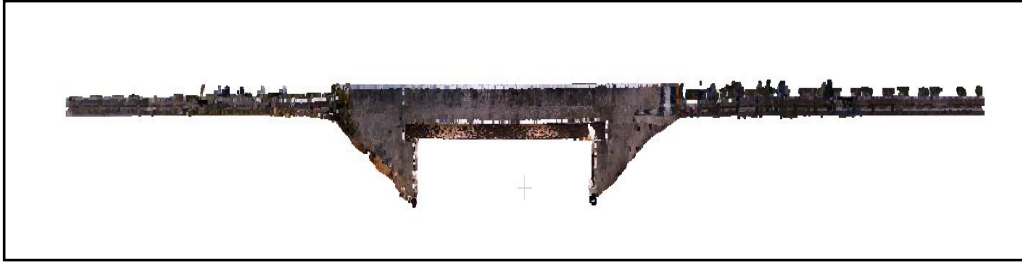
➤ **Bridge Details**

- **Bridge No:** 1040 / 2
- **Chainage:** 1039 + 350
- **Structure type:** MNB

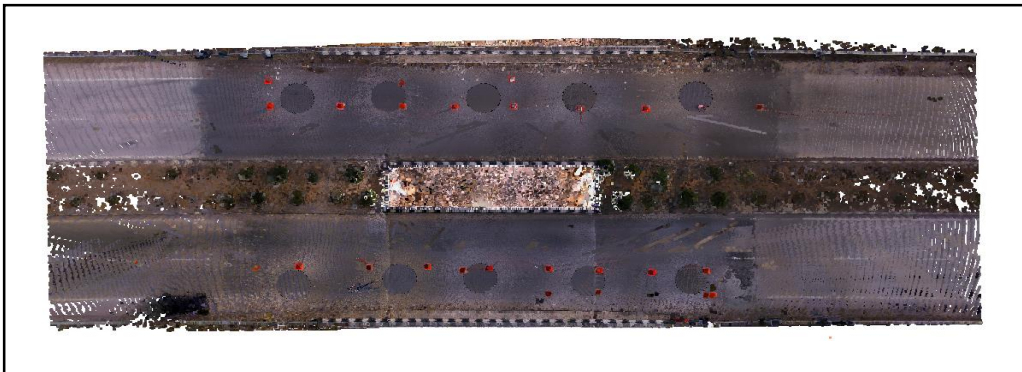
➤ **Project Details**

- **CKRP:** Chittorgarh to kota
- **State:** Rajasthan NH 27
- **Zone:** RJ-06

LiDAR View:



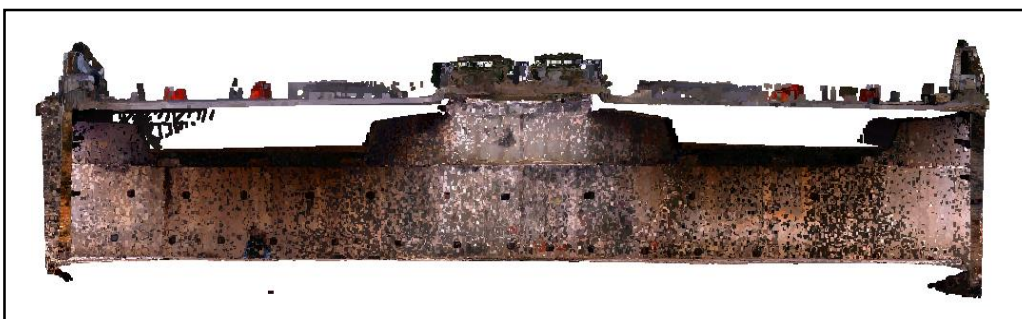
Side View



Top View



Bottom View



Front View

Structural Measurements

1. Height

Measurement	Value
Ground level to Slab bottom	2.60 m
Abutment bottom to Road	4.00 m

2. Length

	Abutment to Abutment	Span 1
LHS	13.92 m	13.92 m
RHS	13.92 m	13.92 m

Deck Width (LHS): 9.89 m

Deck Width (RHS): 9.89 m

Technical Scope of LiDAR Inspection

Category	Details
1. Accuracy Setup and Scan Planning	
Equipment Deployed	Trimble X9 – High-performance terrestrial laser scanner with automatic calibration, IP65-rated rugged build, and in-field registration capability
Point Cloud Density	Configured for high-density capture with sub-centimeter resolution suitable for fine structural detailing
Measurement Accuracy	±1 mm at close range (<60 m); optimal for bridge spans, elevated members, and fascia with scan range up to 150 meters
Coverage Strategy	Comprehensive planning to prioritize critical structural components including abutments, piers, bearings, and deck; ensured overlap to minimize occlusion
2. Field Setup and Scanning Approach	
Site Configuration	Multiple scan positions arranged for 360° coverage including underside views; referenced globally using GNSS/RTK and Ground Control Points (GCPs)
Execution Environment	Scans executed during low-traffic, clear weather conditions to minimize external noise and vibration-related scan distortion
Safety Measures	Operational safety ensured by advance site clearance, barricading, and coordination with local authorities for traffic management
3. Data Resolution and Quality Control	
Resolution Application	Utilized Trimble X9's high-resolution scan settings for capturing large-span structures, vertical components, and minor surface irregularities
Gap Management	Conducted supplementary scans from alternate angles; point cloud refinement and merging handled in Trimble RealWorks and Trimble Perspective
Data Validation	Outlier filtering and noise reduction applied; registration error maintained <5 mm RMSE for consistent dimensional reliability

Equipment Specifications:

Feature	Trimble X9
Scan Speed	1,000,000 pts/sec
Time to Complete a Full Scan	4-6 minutes per setup (depending on resolution)
Point Cloud Accuracy	±3 mm at 20m
Maximum Scanning Range	150m
Field Workflow	Auto-calibration
Imaging	Standard HDR
Durability	IP55

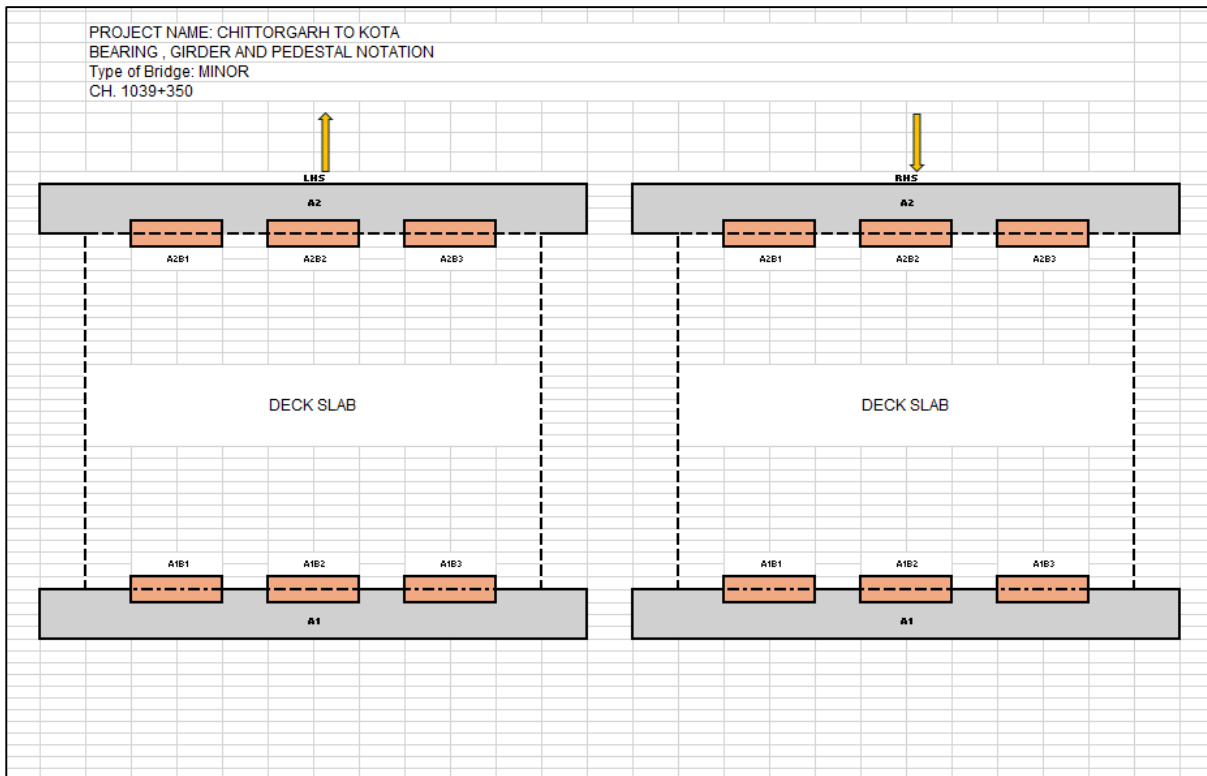
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Phase 1 – Scope of Deliverables

Work Stage	Scope / Deliverables	Location in Report
Accuracy Setup & Scan Planning	Specification of accuracy levels, point cloud density, and scan coverage	Technical Scope of LiDAR Inspection
Technical Setup	Preparation for LiDAR equipment deployment	Technical Scope of LiDAR Inspection
Objective Framing	Component prioritization for scanning	Structural Measurements + Observation Sheet
Field Data Collection	LiDAR data acquisition	Structure Layout + Photos
Resolution Analysis	Data density and resolution analysis	Technical Scope of LiDAR Inspection
Gap Filling	Quality enhancement of point cloud	Embedded in Visual/Photo Section
Point Cloud Analysis	Structural details and interpretation	Observation Sheet + Distress Photos

1. Structure Layout



2. Layout Legends

S. No	Description	Notation
1	Abutment	AB
2	Foundation of Abutment	AF
3	Abutment Cap	AC
4	Bearing Pedestal	BP
5	Bearings	BR
6	Foundation of Pier	PF
7	Pier	PR
8	Pier Cap	PC
9	Span	SP
10	Slab span	SL
11	Box Girder span	BG
12	I Girder	IG
13	Rectangular Girder	RG
14	Square Girder	SG
15	T Beam Girders	TB
16	Concrete Deck	CC
17	Bituminous Deck	BD
18	Slab b/n girders	BS
19	Cross Girders	CG
20	Railing LHS	RL
21	Railing RHS	RR
22	Footpath LHS	FL
23	Footpath RHS	RR
24	Expansion Joints	EJ
25	Pourable/Crack Joint Seal	PJ
26	Compression Joint Seal	CJ
27	Assembly Joint Seal	AJ
28	Strip Seal Joint	SJ
29	Drainage Spouts	DS
30	Wing Wall	WW
31	Reinforced earth wall	RE
32	Approach Slab	AS

3. Distress Nomenclature

Legend	Description	Legend	Description
BG	Bulging	SC	Steel Corroded
CRA	Map Crack/ Crazing	SE	Steel Exposed
RC	Crack along main reinforcement	SPE	Seepage Evident
RP	Crack perpendicular to Main Reinforcement	SPM	Seepage Marks
MC	Minor Crack	SL	Scaling
WC	Wide Crack	SG/(G)(S)	Sag in Girder, Slab
SPL	Spalling	EFF	Efflorescence
DL	Delaminated Section	STL	Settlement
HC	Honey Combing	H	Hollow Section
TL	Tilt	VEG	Vegetation Growth
		PD	Pedestal Damage

4. Observations

LHS:

Element Name	Observation	Location of Distress	Photo No.	Units	Distress Details			Area of Repair (Sq./RM T/m3)
					L(M)	B(M)	D(M)	
Abutment 1	Hollow Section		1		1.32	0.71	0.063	0.0590
Dirt Wall	Steel Exposed		2		0.43	0.24		0.1032
Abutment 1	Hollow Pocket		3		0.85	0.52	0.024	0.0106
Abutment 1	Hollow Pocket		4		0.27	0.20	0.055	0.00297
Abutment 1	Steel Exposed		5		0.15	0.15		0.0225
Abutment 1	Hollow Pocket		6		1.53	0.46	0.045	0.03167
Dirt Wall	Honey combing		7		0.22	0.19		0.0418
Dirt Wall	Steel Exposed		8		0.84	0.53		0.4452
Abutment 2	Hollow Pocket		9		0.50	0.42	0.045	0.00945
Abutment 2	Scaling		10		0.48	0.41		0.1968
Abutment 2	Hollow Pocket		11		0.74	0.41	0.052	0.01577
Abutment 2	Scaling		12		0.87	0.55		0.4785

RHS:

Element Name	Observation	Location of Distress	Photo No.	Units	Distress Details			Area of Repair (Sq./RM T)
					L(M)	B(M)	D(M)	
Abutment 1	Hollow Pocket		13		0.36	0.29	0.048	0.00501
Abutment 1	Spalling		14		2.12	0.43		0.9116
Abutment 1	Spalling		15		3.67	0.61		2.2387

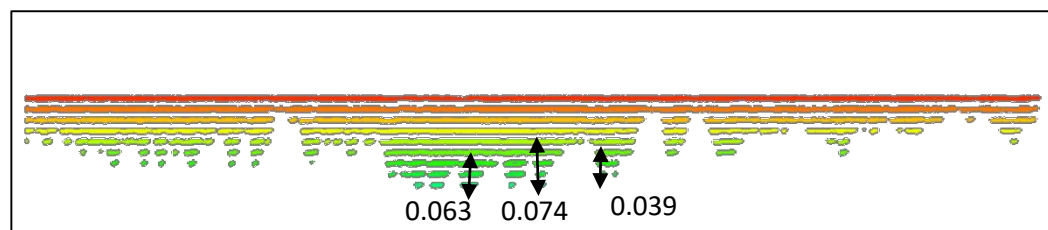
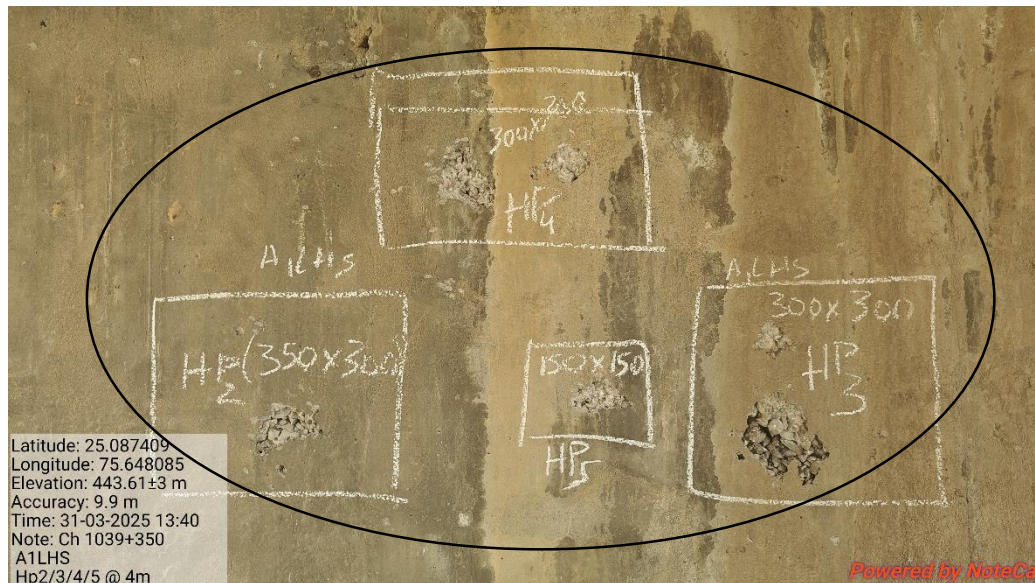
Observation sheet Non- Structural Elements

Sl. No.	Description	Distress Notation	Observation	Location		Distress Details			Area of Repair
				X (m)	Y (m)	L (m)	W (m)	D (m)	Sqm
1	Return wall		OK						
2	Riding return wall		NA						
3	Wing wall		NA						
4	Butterfly walls at top of median wall		NA						
5	Dirt wall at top of abutment		Ok						
6	Gabion walls		NA						
7	Hazard Marker sign board		OK						
8	Hand railing		NA						
9	Crash barrier		OK						
10	Median wall		NA						
11	Pipe Hand Railing (on crash Barrier)		OK						
12	Cleaning of expansion joints		Choked						
13	Wheel guard (Mounted & Unmounted)		NA						
14	Footpath		OK						
15	Drainage Spouts		Choked						
16	Drainage Spout Grating		Choked						
17	Drainage Spout Extension Pipe		OK						
18	Weep Holes		NA						
19	Fascia Panel		NA						

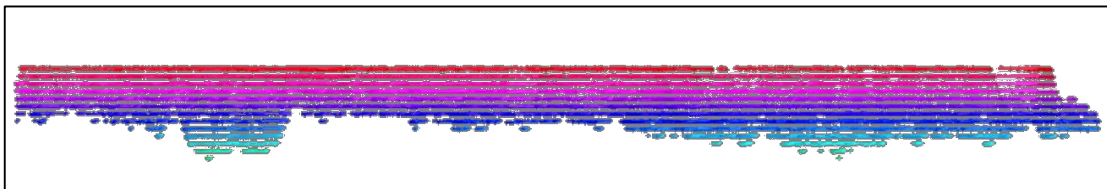
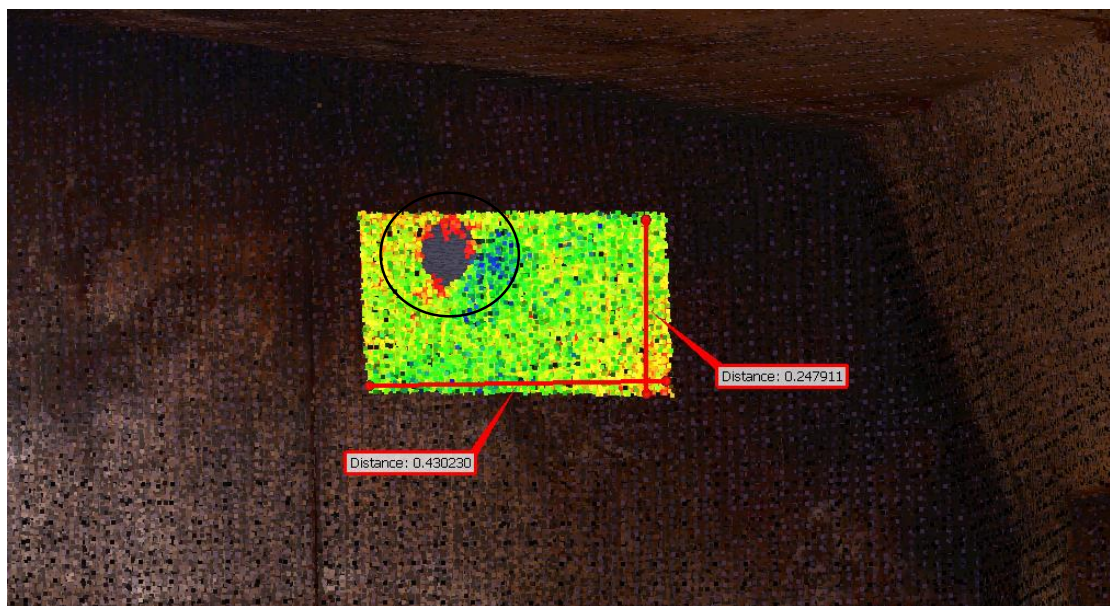
Sl. No.	Description	Distress Notation	Observation	Location		Distress Details			Area of Repair
				X (m)	Y (m)	L (m)	W (m)	D (m)	Sqm
20	Friction Slab		Ok						
21	Approach slab including PCC		OK						
22	Backfilling below Approach slab		OK						
23	Stone Pitching in quarter coning		NA						
24	Filling in quarter coning		NA						
25	Rigid apron		NA						
26	Curtain wall		NA						
27	Flexible apron		NA						
28	Vent clearance		NA						
29	Toe wall of Quarter coning		NA						
30	RCC drains in tunnel		NA						
31	Cover slab of RCC drains		NA						
32	RCC portal of tunnel		NA						
33	Counter fort wall		NA						
34	Booths structure		NA						
35	RCC Booth guard		NA						
36	C.C Panels including PCC		NA						
37	Pipe drain		NA						
38	Pipe drain cover		NA						
39	Drain openings		NA						

CH: 1039+350 – MNB

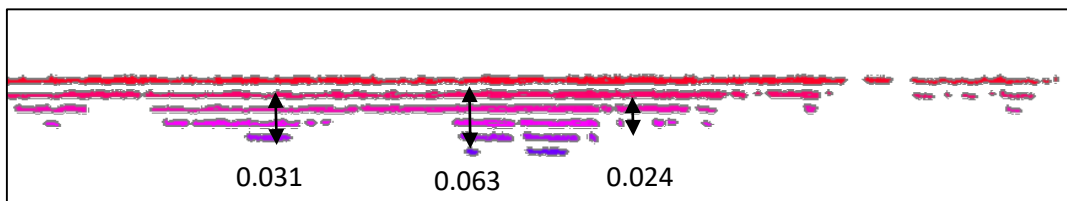
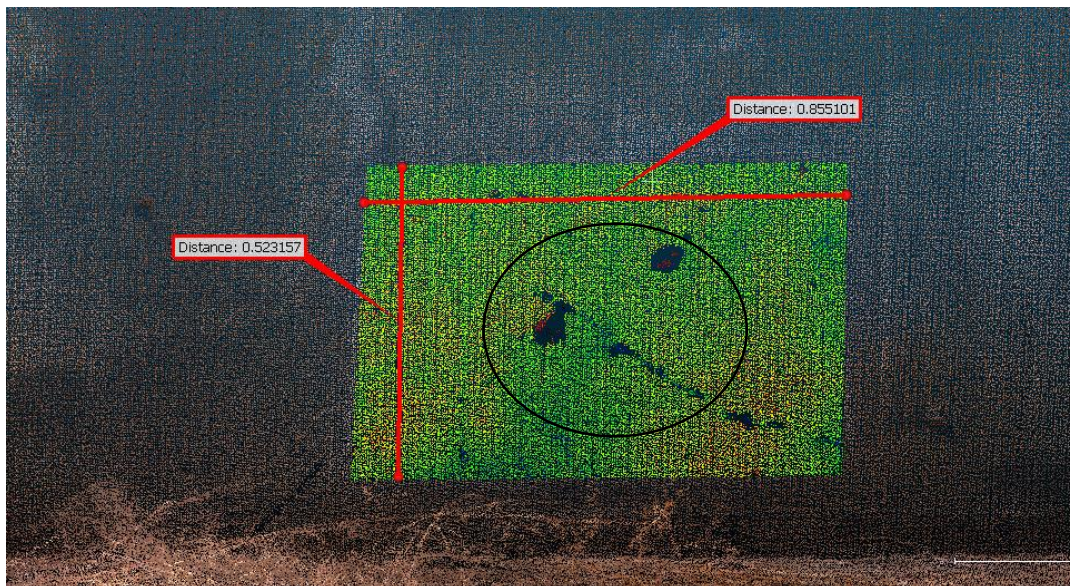
DISTRESS PHOTOS



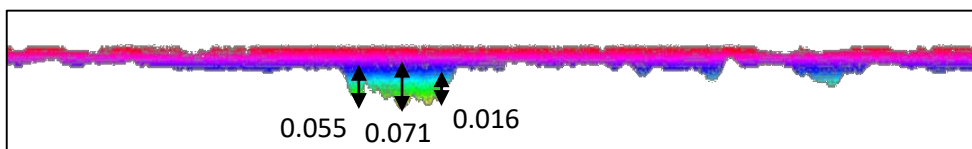
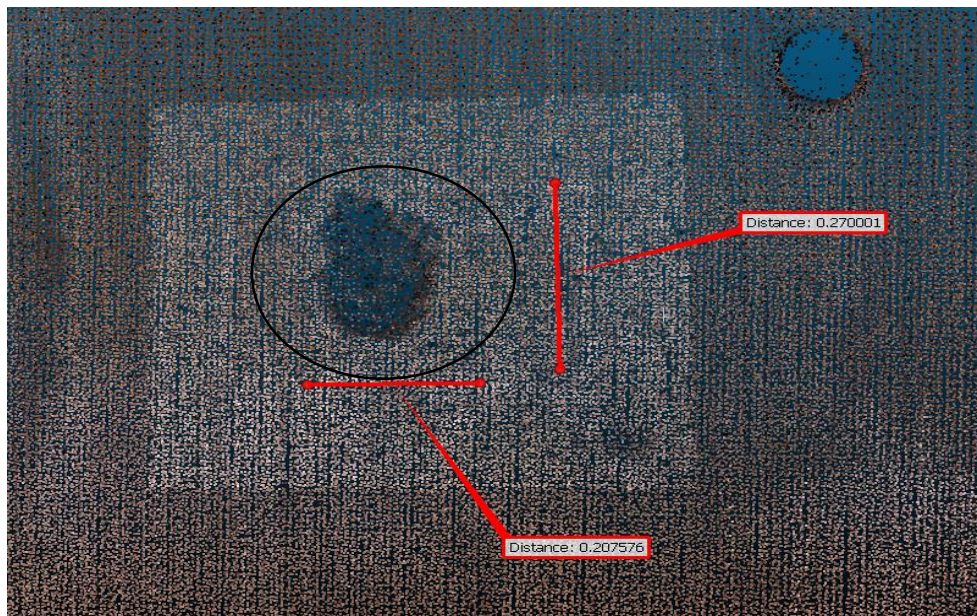
1. Hollow section observed on LHS Abutment 1, measuring 1.32 m in length and 0.71 m in width, covering an area of 0.9372 m². Depth ranges from 0.039 m (min) to 0.074 m (max), with an average depth of 0.063 m. The calculated volume of the distress is 0.0590 m³.



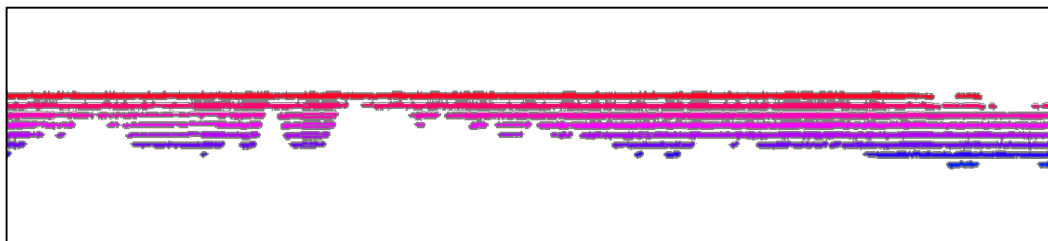
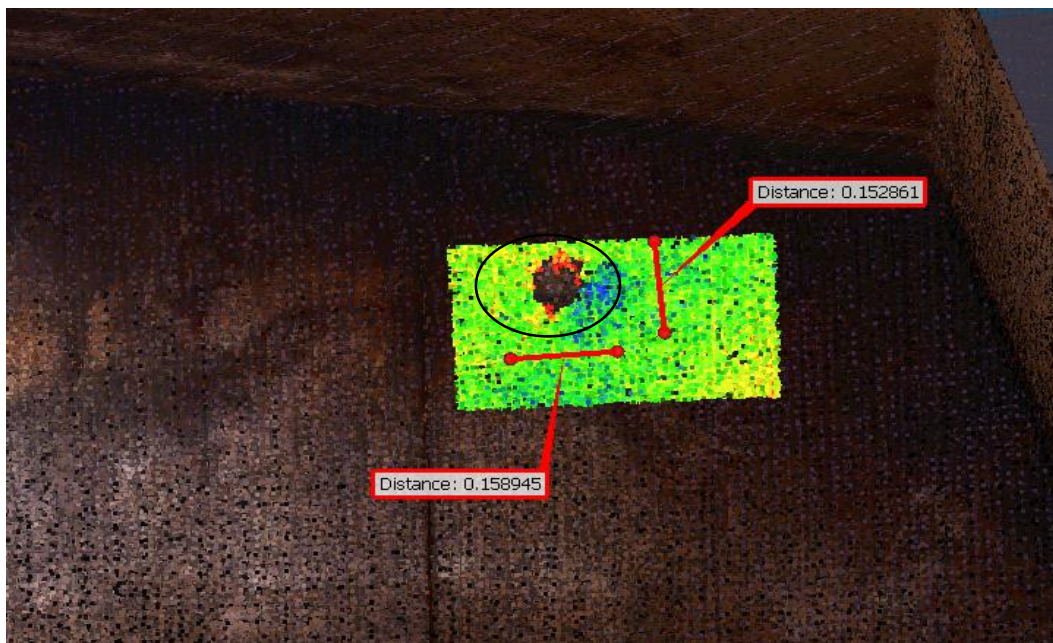
2. Steel exposure with dirt accumulation observed on LHS Abutment 1 wall, measuring 0.43 m in length and 0.24 m in width, area 0.1032 m²



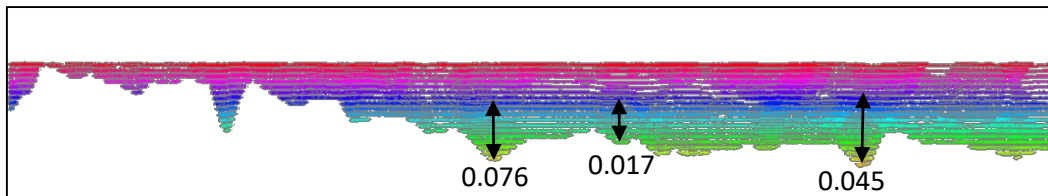
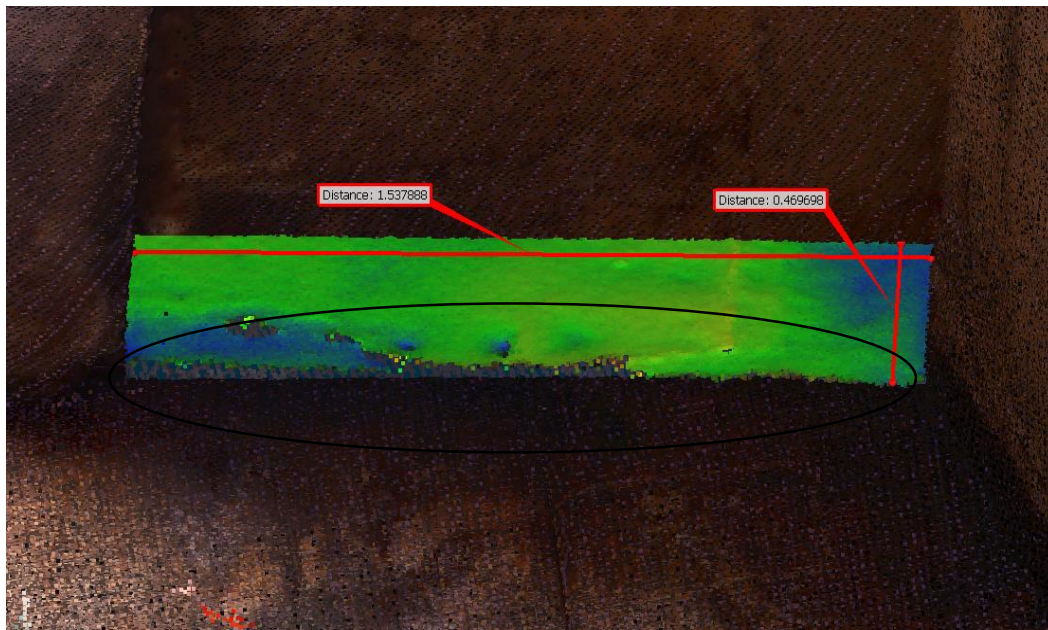
3. Hollow pocket observed at on LHS Abutment 1, measuring 0.85 m in length and 0.52 m in width, Depth ranges from 0.031 m (min) to 0.063 m (max), with an average depth of 0.024 m. The calculated volume of the distress is 0.0106 m³.



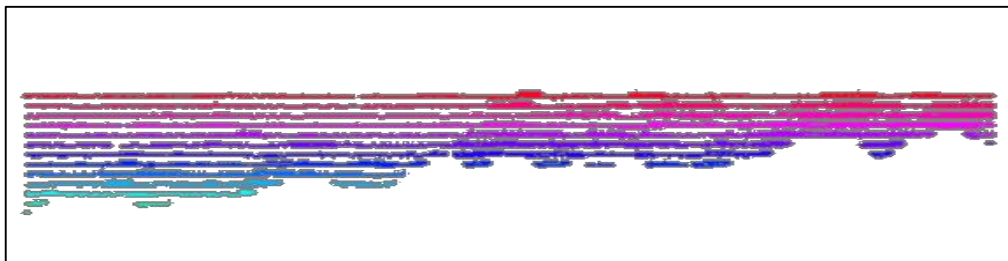
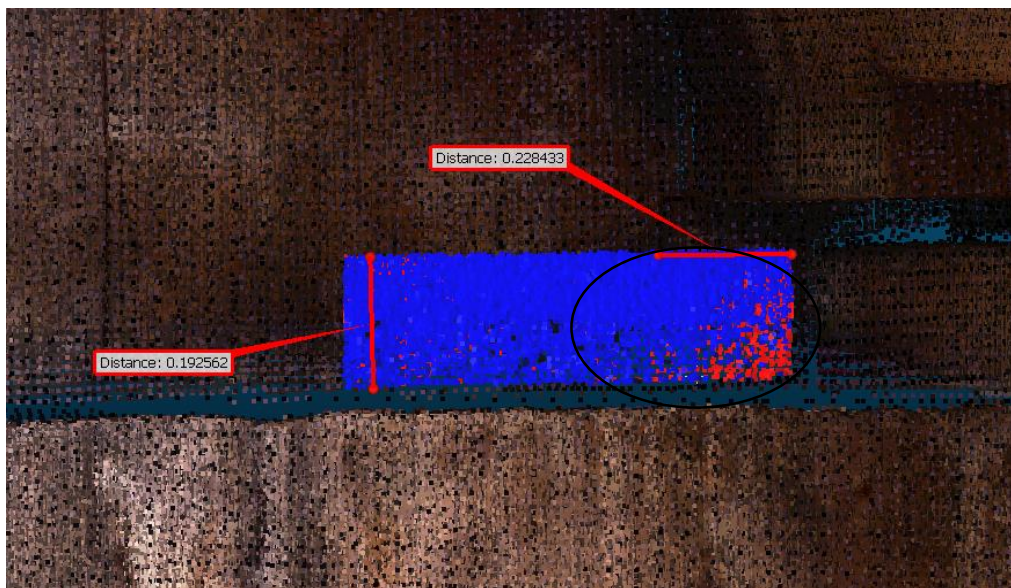
4. Hollow pocket observed on LHS Abutment 1, measuring 0.27 m in length and 0.20 m in width, Depth ranges from 0.016 m (min) to 0.071 m (max), with an average depth of 0.055 m. The calculated volume of the distress is 0.00297 m³.



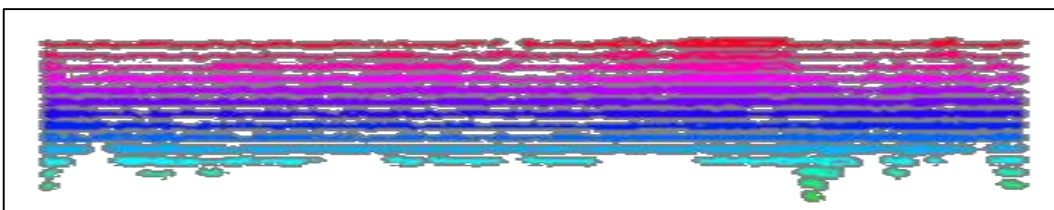
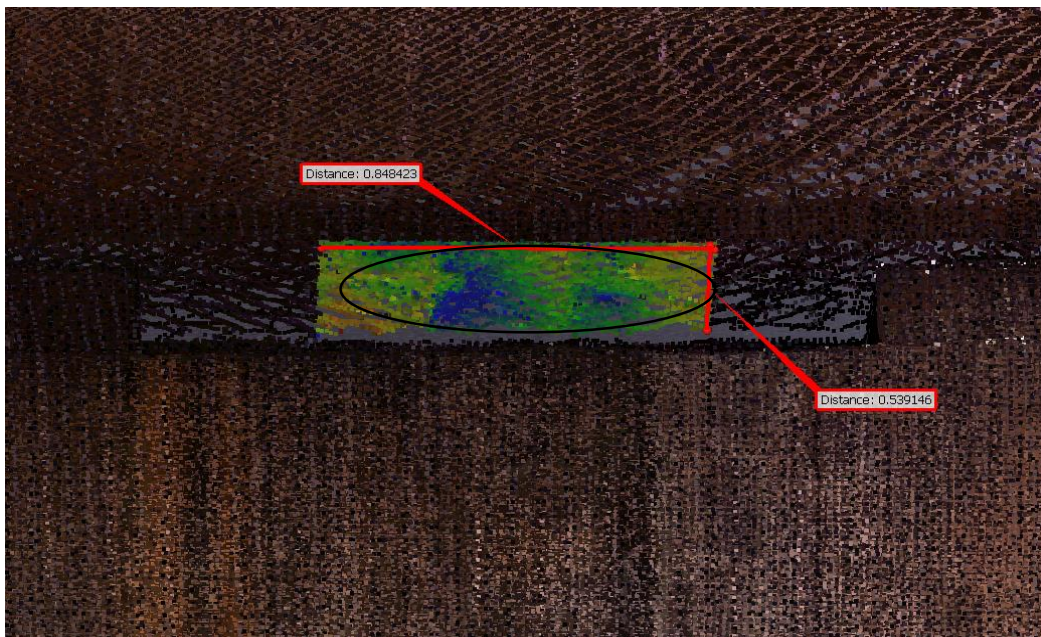
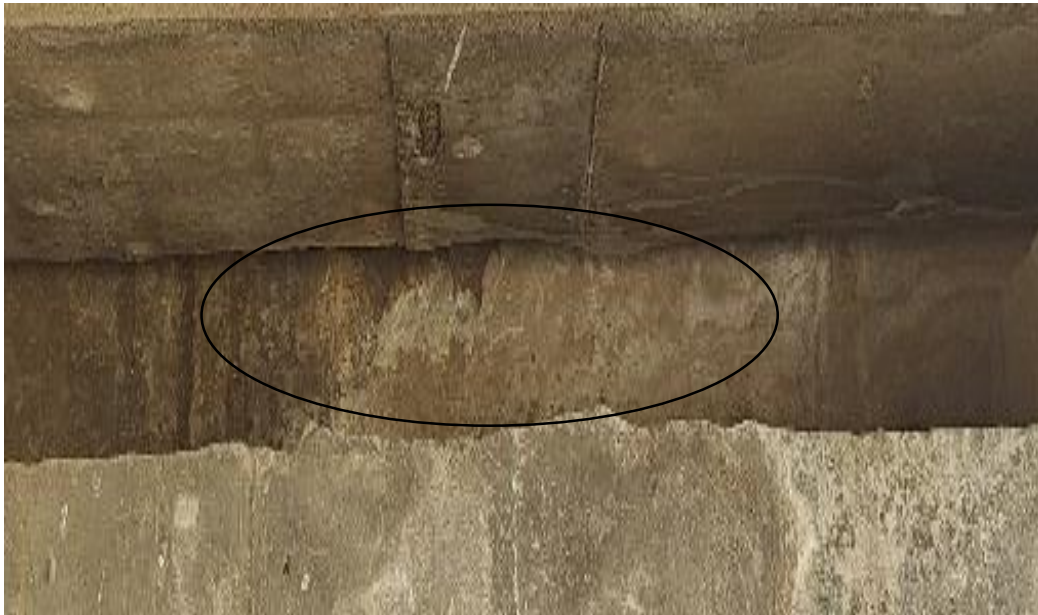
5. Steel exposed surface observed on LHS Abutment 1, measuring 0.15 m in length and 0.15 m in width, covering 0.0225 m².



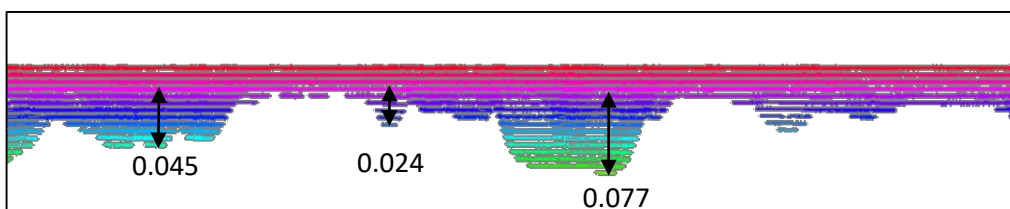
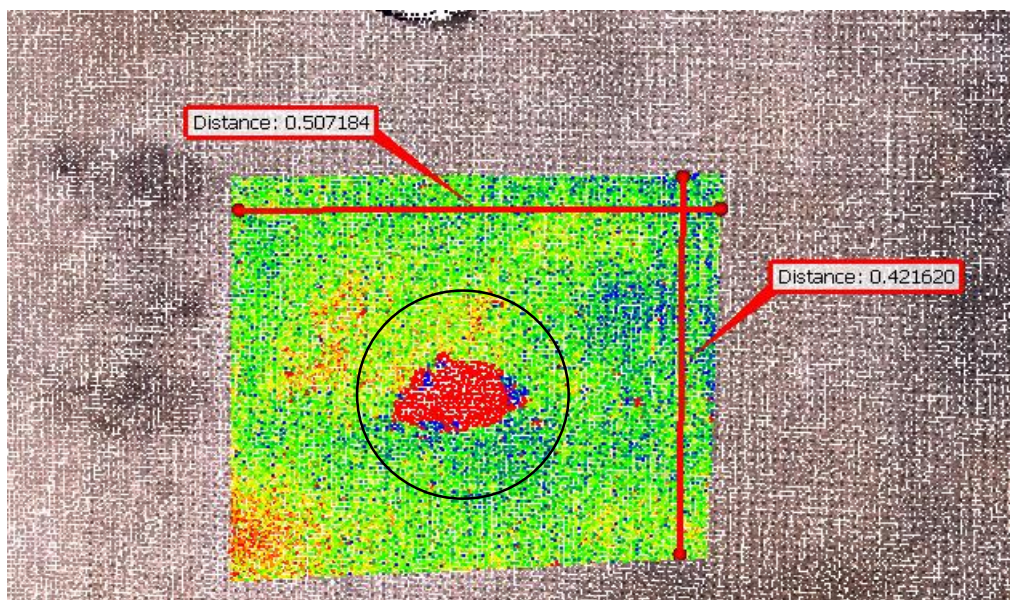
6. Hollow pocket observed on approach slab dirt wall bottom, LHS Abutment 1, measuring 1.53 m in length and 0.46 m in width, Depth ranges from 0.016 m (min) to 0.071 m (max), with an average depth of 0.045 m. The calculated volume of the distress is 0.03167 m³.



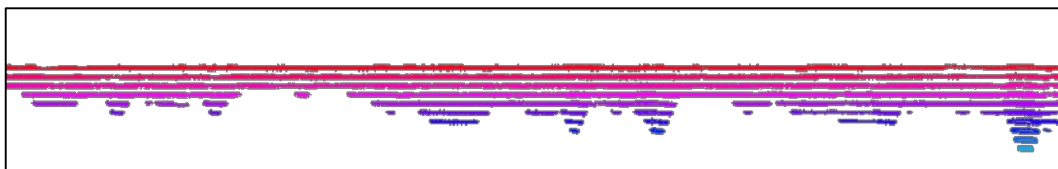
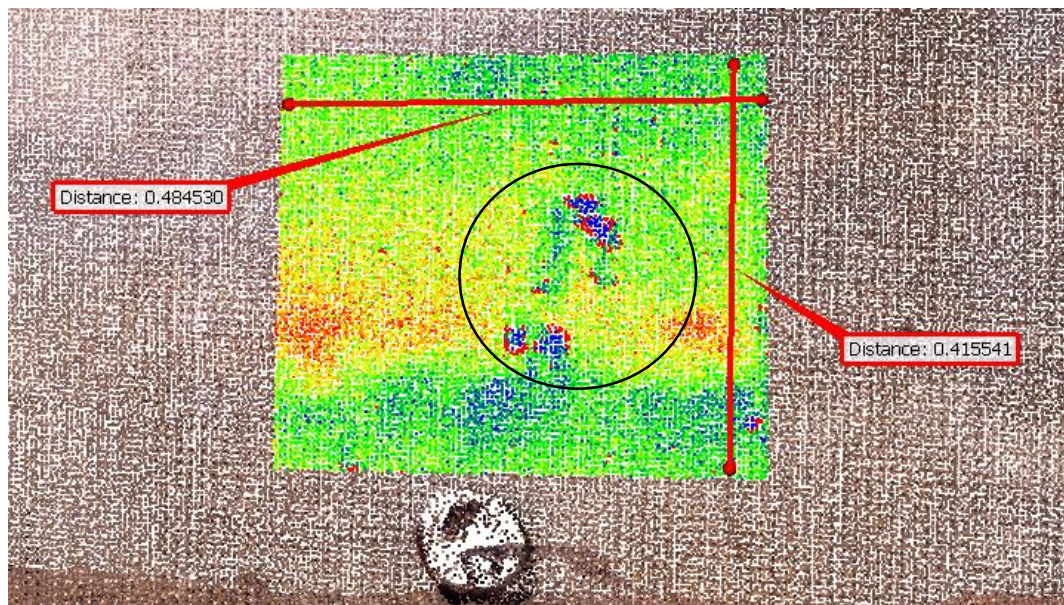
7. Honeycombing observed at Bearing 1 pedestal dirt wall, LHS Abutment 1, measuring 0.22 m in length and 0.19 m in width, area 0.0418 m².



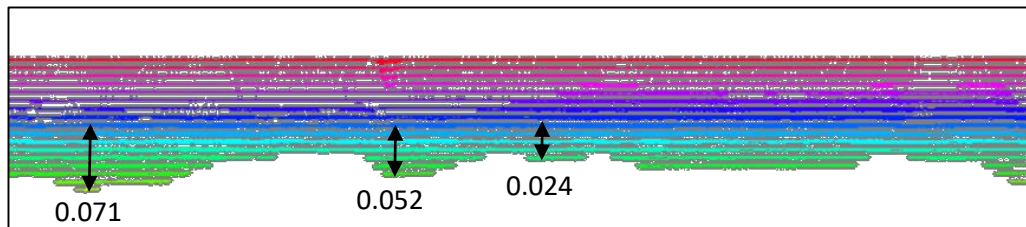
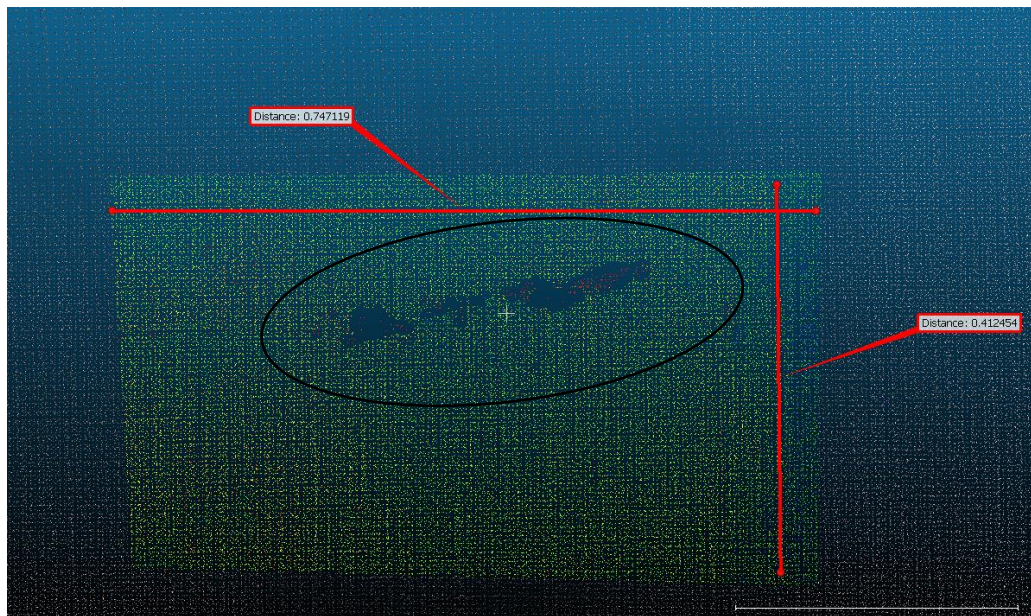
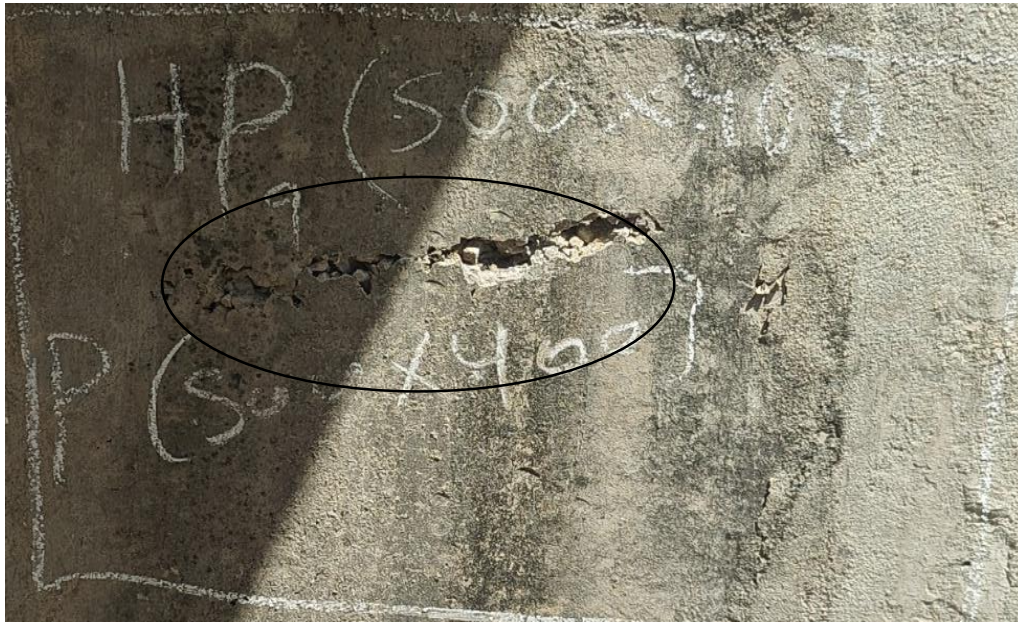
8. Steel exposed section observed between Bearing 2–Bearing 3, LHS Abutment 1, measuring 0.84 m in length and 0.53 m in width, covering 0.4452 m².



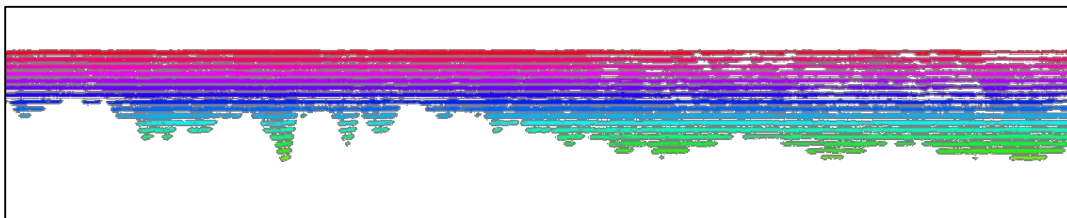
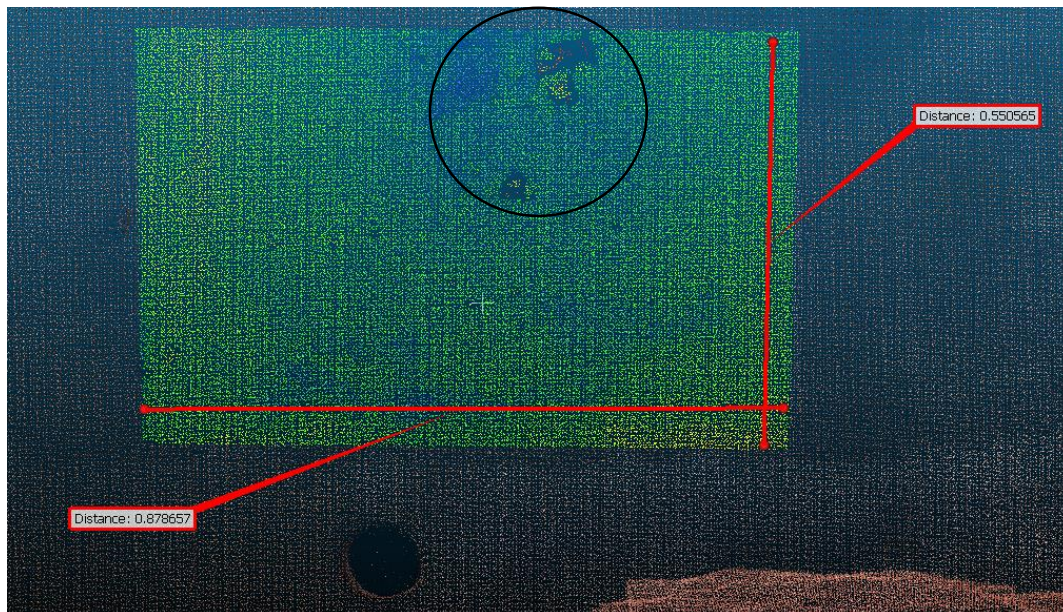
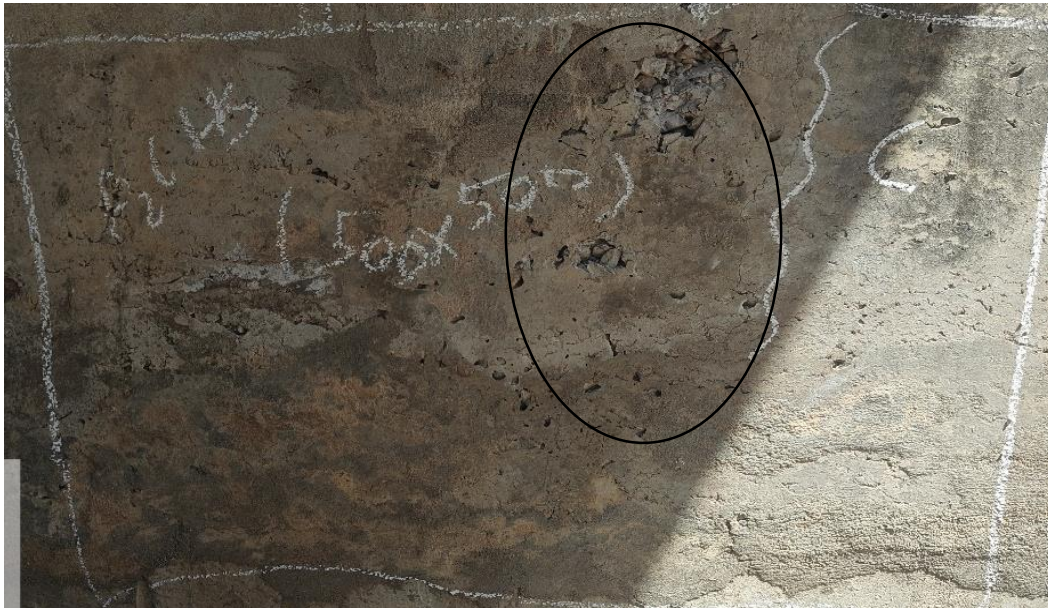
9. Hollow pocket observed on LHS Abutment 2 , measuring 0.50 m in length and 0.42 m in width, covering 0.21 m². Depth ranges from 0.024 m (min) to 0.077 m (max), with an average depth of 0.045 m. The calculated volume of the distress is 0.00945 m³.



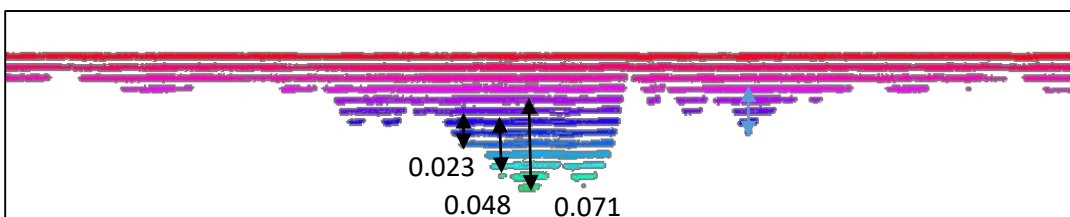
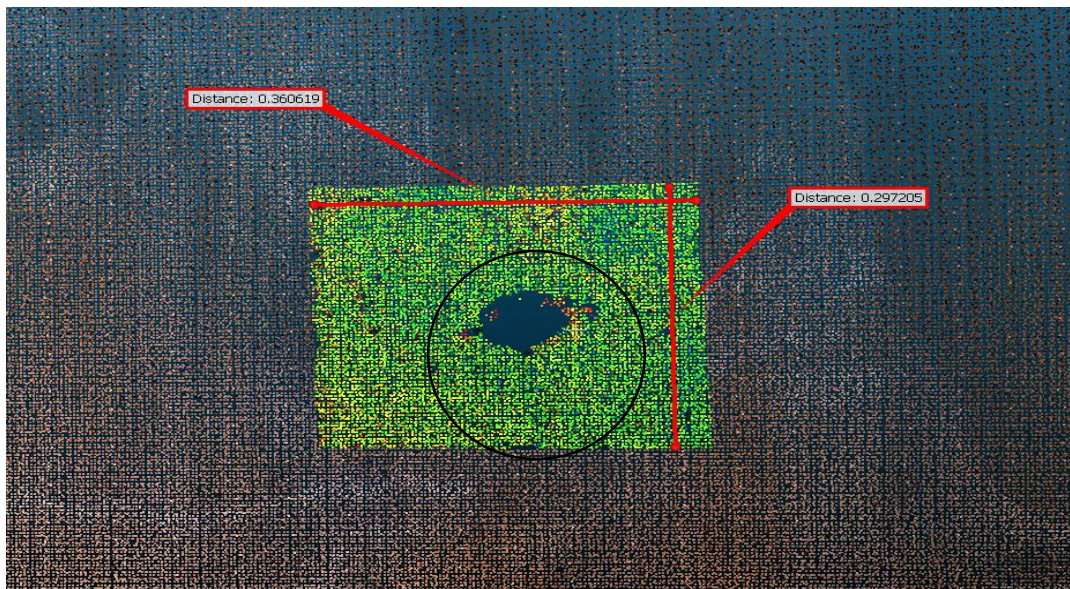
10. Scaling observed on LHS Abutment 2, measuring 0.48 m in length and 0.41 m in width, area 0.1968 m²



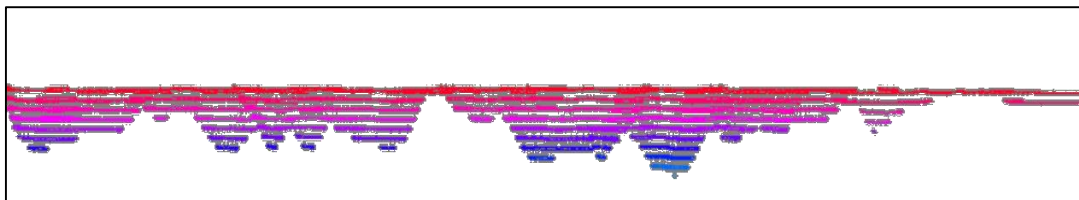
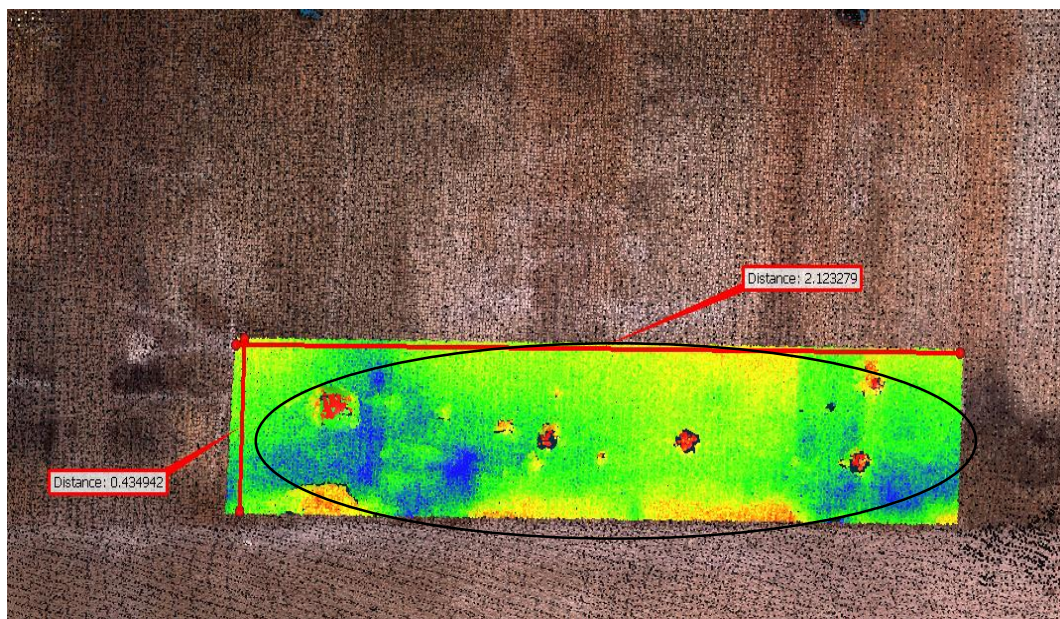
11. Hollow pocket observed on LHS Abutment 2, measuring 0.74 m in length and 0.41 m in width, Depth ranges from 0.024 m (min) to 0.071 m (max), with an average depth of 0.052 m. The calculated volume of the distress is 0.01577 m³.



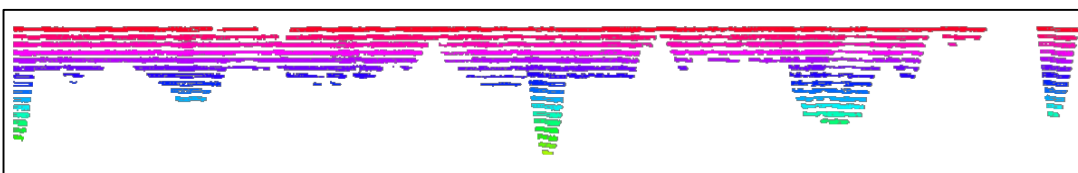
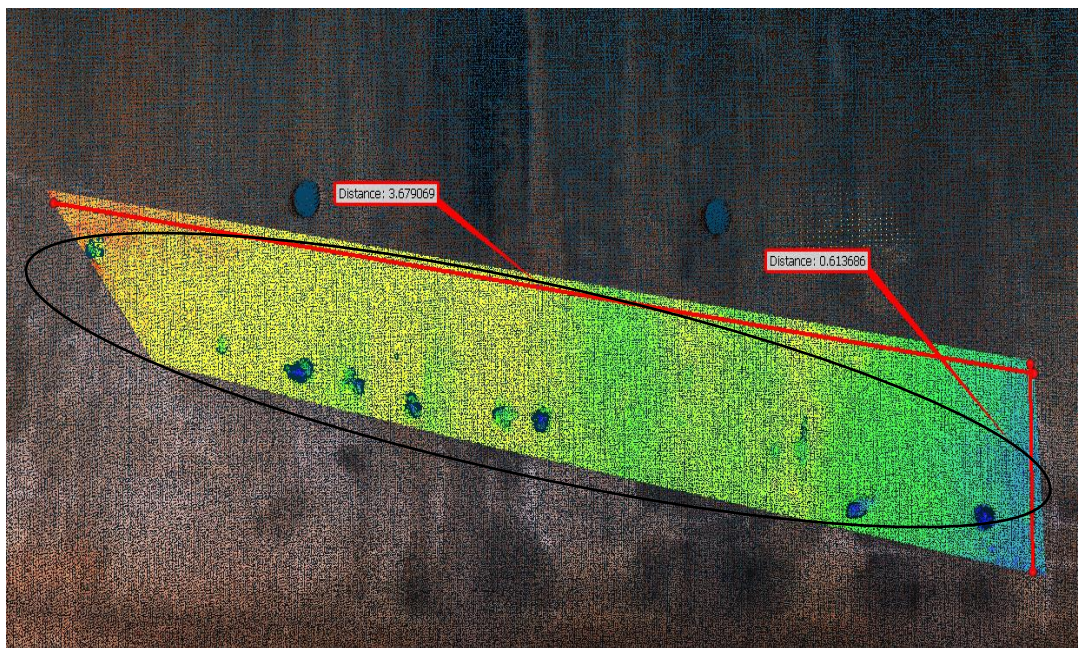
12. Scaling observed on LHS Abutment 2, measuring 0.87 m in length and 0.55 m in width, area 0.4785 m².



13. Hollow pocket observed on RHS Abutment 1 measuring 0.36 m in length and 0.29 m in width, Depth ranges from 0.023m (min) to 0.071 m (max), with an average depth of 0.048 m. The calculated volume of the distress is 0.00501m³.



14. Spalling observed on RHS Abutment 1 due to sulphate attack, measuring 2.12 m in length and 0.43 m in width, covering an area of 0.9116 m².



15. Spalling observed on RHS Abutment 1 due to sulphate attack, measuring 3.67 m in length and 0.61 m in width, covering an area of 2.2387 m².